Reg. No. | | | | | | | | | | | [14MAE31]

THIRD SEMESTER B.TECH EXAMINATION, DECEMBER–2015

MATHEMATICS-III

Time:3 Hours **Maximum Marks:70**

Instructions: Answer five full questions, choosing at least TWO from each part.

PART - A

1. a. State and prove Cauchy's mean value theorem. **05**

   b. Prove that: $\sqrt{1+\sin 2x} = 1 + x - \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24} + \ldots\ldots$. **05**

   c. Evaluate $\underset{x\to a}{Lt}[\{2-(x/a)\}]^{\tan(\pi x/2a)}$. **04**

2. a. If $u = e^{xyz}$, find the vlaue of $\frac{\partial^3 u}{\partial x \partial y \partial z}$. **05**

   b. If $u = f(r)$ where $r=\sqrt{x^2+y^2}$ then show that $u_{xx} + u_{yy} = f''(r) + (1/r) f'(r)$. **05**

   c. State and prove Euler's theorem **04**

3. a. If $u = x\log(xy)$ where $x^3 + y^3 + 3xy = 1$, *find* $\frac{du}{dx}$ **06**

   b. If $x = u(1-v)$ *and* $y = uv$ then prove that $J.\ J^1 = 1$ **08**

4. a. Expand $x^2y + 3y - 2$ in powers of (x-1) and (y+2) using Taylor's series expansion. **05**

   b. Examine the function $f(x, y) = x^3 + y^3 - 3axy$, a>O for extreme values. **05**

   c. Find the minimum value of $x^2+y^2+z^2$ when x + y + z = 3a **05**

PART -B

5. a. Expand $f(x)=x^2$ as a Fourier's series in the internal $(-\pi, \pi)$ **05**

   b. Find the half range sine series for the function $f(x)=(x-1)^2$ in the interval (0,1). **04**

   c. Compute the Fourier series up-to first harmonic for the first harmonic for the following values: **05**

X	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π
F(x0	1.98	1.30	1.05	1.30	-0.88	-0.25	1.98

6. a. Find the Fourier's transform of $f(x) = \begin{cases} (1-x^2), & |x|<1 \\ 0, & |x|>1 \end{cases}$ **05**

   b. Find the Fourier's sine transform of $\frac{e^{-ax}}{x}, a > 0$ **05**

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c. Find the Fourier cosine transform of $f(x)=\begin{cases} x, 0<x<1 \\ 2-x, 1<x<2 \\ 0, x>2 \end{cases}$ **05**

7. **a.** Derive a partial differential equation from the function Z=f(x+at)+g(x-at) **05**

b. Solve: $\frac{\partial^3 z}{\partial x^2 \partial y}+18xy^2+\sin(2x-y)=0$. **04**

c. Solve : (mz - ny) P+(nx - lz) q = (ly - mx). **05**

8. **a.** Solve the wave equation $\frac{\partial^2 u}{\partial t^2}=c^2\frac{\partial^2 u}{\partial x^2}$ given that u(0, t) = 0, u (l, t)=0, $\frac{\partial u}{\partial t}=0$ when t=0 and u (x, 0) $=u_o \sin\left(\pi x/l\right)$. **07**

b. Solve the differential equation $\frac{\partial u}{\partial t}=\alpha^2\frac{\partial^2 u}{\partial x^2}$ for the conduction of heat along a rod without radiation, subject to the following conditions. **07**

i. u is not infinite for $t \to \infty$

ii. $\frac{\partial u}{\partial x}=0$ for x = 0 and x=l

iii. U = lx – x^2 for t=0, between x = 0 and x =l